Calibrating the Coupon Strike of an Autocallable Note under Local Volatility *

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1 Introduction

Autocallable notes are popular market-linked structured products that offer enhanced coupon yields alongside conditional capital protection. These notes have early redemption features triggered when the underlying price reaches a predefined knock-out level on scheduled observation dates. This project studies a multi-asset case, an autocallable note tied to a basket of three equity indices (Nikkei 225, S&P 500 and Hang Seng Index), where the coupon, the knock-out and the knock-in all depend on the worst performer. The objective is to price the note by Monte Carlo simulation under a local volatility surface, and to calibrate the coupon strike at which the note prices at 98% of its issue price. The remainder of the report is organized as follows. Section 2 sets out the note structure, and Section 3 describes the data. Section 4 fits and selects the volatility surface. Section 5 specifies the simulation and pricing. Section 6 calibrates the coupon strike. Section 7 concludes.

2 Note Structure

The note is written on the underlying basket of the Nikkei 225 (NKY), the S&P 500 (SPX) and the Hang Seng Index (HSI). It is issued at par and runs for two years from a trade date of 11 November 2023. Throughout, k indexes the three indices, S_t^k denotes the closing level of index k at time t and S_0^k its initial level.

Note denomination, N	USD 10,000
Issue price	100% of N
Trade date	11 November 2023
Maturity date	2 years after trade date
Observation dates, t_m	every six months, $m = 1, \ldots, 4$
Maximum rate of interest, r_+	2% per annum
Minimum rate of interest, r_-	0.01% per annum
Knock-out level, B_+	1.10 (110% of initial spot)
Knock-in level, B_-	0.50 (50% of initial spot)

The variable interest reference price is the relative level of the laggard underlying, $R_t = \min_k S_t^k / S_0^k$. At each observation date t_m, the note pays a coupon rate of r_+ if the reference price is at or above the coupon strike c, and r_- otherwise. The coupon strike is the parameter that the project calibrates. A knock-out event occurs when the reference price reaches a threshold of B_+ on an observation date. In this case, the note redeems for its full denomination and expires, and the coupon for that period is still paid. A knock-in event occurs if the reference price falls to or below the threshold B_- at any time during the life of the note. At the final redemption, if no knock-in event has occurred, the note repays the full denomination, and if one has occurred, the note repays $N \cdot \min(1, R_T)$.

*Code and replication data are available at *GitHub*.

3 Data

Option chains for the three indices were fetched from Bloomberg as of 11 November 2023. Entries without a last price were dropped. Overnight index swap rates were quoted on a daily compounding basis. On a day-count basis β (365 for Hong Kong and Japan, 360 for the United States), the quoted rate r_{daily} is converted to a continuously compounded rate by $r = \beta \ln\left(1 + \frac{r_{\text{daily}}}{\beta}\right)$, and the resulting rates are interpolated across maturity by a cubic spline to give $r(T)$.

3.1 At-the-money forward

Put-call parity under a continuous dividend yield q relates the call price C and the put price P at strike K and maturity T by $C(K) - P(K) = S_0 e^{-qT} - Ke^{-rT}$. Setting $C = P$ gives

$$K_{\text{ATMF}} = S_0 e^{(r-q)T}.$$

Since the strike is not quoted under the discrete chain, we locate the tightest interval $[K_1, K_2]$ s.t.

$$C(K_1) > P(K_1) \quad \text{and} \quad C(K_2) < P(K_2),$$

which brackets K_{ATMF}. We then approximate the first derivatives at K_1 by finite differences, where Φ is the standard normal distribution function and d_2 the second Black-Scholes term.

$$\begin{aligned}
\frac{C(K_{\text{ATMF}}) - C(K_1)}{K_{\text{ATMF}} - K_1} &\approx \left.\frac{\partial C}{\partial K}\right|_{K_1} \approx -e^{-rT}\Phi(d_2)\big|_{K_1}, \\
\frac{P(K_{\text{ATMF}}) - P(K_1)}{K_{\text{ATMF}} - K_1} &\approx \left.\frac{\partial P}{\partial K}\right|_{K_1} \approx \quad e^{-rT}\Phi(-d_2)\big|_{K_1}, \\
\frac{C(K_{\text{ATMF}}) - C(K_1)}{K_{\text{ATMF}} - K_1} &\approx \left.\frac{\partial C}{\partial K}\right|_{K_1} \approx \frac{C(K_2) - C(K_1)}{K_2 - K_1}.
\end{aligned}$$

Near the forward, $\Phi(-d_2) \approx \Phi(d_2) \approx \frac{1}{2}$, so the two derivatives are approximately equal and opposite. Therefore, $C(K_{\text{ATMF}}) = P(K_{\text{ATMF}})$ gives

$$C(K_{\text{ATMF}}) \approx \frac{C(K_1) + P(K_1)}{2},$$

$$K_{\text{ATMF}} \approx \frac{C(K_{\text{ATMF}}) - C(K_1)}{C(K_2) - C(K_1)} (K_2 - K_1) + K_1. \tag{1}$$

3.2 Implied volatility and dividend yield

The implied variance σ^2_{BS} is recovered from the out-of-the-money quote at each strike by bisection on the Black-Scholes price, described in Appendix A. The implied dividend yield q follows from the forward strike relation under a non-negativity floor,

$$q \approx \max\left(0,\ r - \frac{\ln(K_{\text{ATMF}}/S_0)}{T}\right),$$

which is interpolated across maturity by a cubic spline to give $q(T)$.

3.3 Local volatility

The local variance follows from the Dupire equation (Dupire, 1994),

$$\sigma^2_{\text{loc}}(K,T) = \frac{\partial C/\partial T}{\frac{1}{2}K^2\,\partial^2 C/\partial K^2}, \tag{2}$$

derived in Appendix C, with both derivatives taken by three-point central differences on the Black-Scholes price in the forward, using a step of one day in maturity and one percent of the strike spacing in strike.

3.4 Gatheral's mapping

As a third route, the fitted implied surface is mapped to local variance through Gatheral (2002). In terms of the total variance $w = \sigma^2_{\text{BS}}T$, the Dupire equation becomes the relation derived in Appendix D,

$$\sigma^2_{\text{loc}} = \frac{\partial w/\partial T}{1 - \frac{y}{w}\frac{\partial w}{\partial y} + \frac{1}{4}\left(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2}\right)\left(\frac{\partial w}{\partial y}\right)^2 + \frac{1}{2}\frac{\partial^2 w}{\partial y^2}}. \tag{3}$$

4 Volatility Surface

4.1 Fitting the surface

At each maturity, the implied and the local variance are each fitted across log forward moneyness $y = \ln(K/K_{\text{ATMF}})$ by

$$\sigma^2(y) = \sigma^2_{\text{ATMF}} + \delta\,\frac{\tanh(\kappa y)}{\kappa} + \frac{\gamma}{2}\left(\frac{\tanh(\kappa y)}{\kappa}\right)^2, \tag{4}$$

where σ^2 stands for either variance, δ is the skew, γ the convexity and κ the celerity.

Curves that are concave are dropped since a concave curve inverts the smile and destabilizes the interpolation across maturity that follows. This removes two implied and five local curves for the Hang Seng Index and one implied and seven local curves for the S&P 500, and none for the Nikkei 225. The fitted curves are shown in Appendix F.

The estimation of a continuous volatility surface involves interpolating from the discrete fitted volatility curves. Here, we explore the linear and cubic spline interpolations. Linear interpolation joins adjacent maturities by a straight line, so that a positive variance at every knot guarantees a positive variance between them. The cubic spline of Appendix B produces a smoother surface, fitting a piecewise cubic whose first and second derivatives are continuous at every knot.

4.2 Selecting the surface

Six candidate surfaces follow, being the implied, the Dupire and the Gatheral variance under each of the two interpolations, and they are plotted in Appendix F. Table 1 reports the percentage of the moneyness window on which the fitted variance is negative. Linear interpolation is never worse than cubic, and is strictly better in six of the nine pairs. Gatheral's mapping is the weakest of the three, leaving between 10.5% and 33.6% of the window with a negative variance, which we attribute to the mapping being applied to an already fitted implied surface rather than to the data, so that estimation error from the first fit propagates into the second. The simulation employs the Dupire local variance under linear interpolation, and Figure 1 shows the fitted surface.

Surface	NKY	HSI	SPX
Implied, cubic	0.0	0.0	10.4
Implied, linear	0.0	0.0	0.0
Dupire, cubic	0.0	12.5	10.6
Dupire, linear	0.0	0.2	0.0
Gatheral, cubic	15.7	14.0	33.6
Gatheral, linear	11.7	10.5	16.1

Table 1: Percentage of the moneyness window on which the fitted variance is negative.

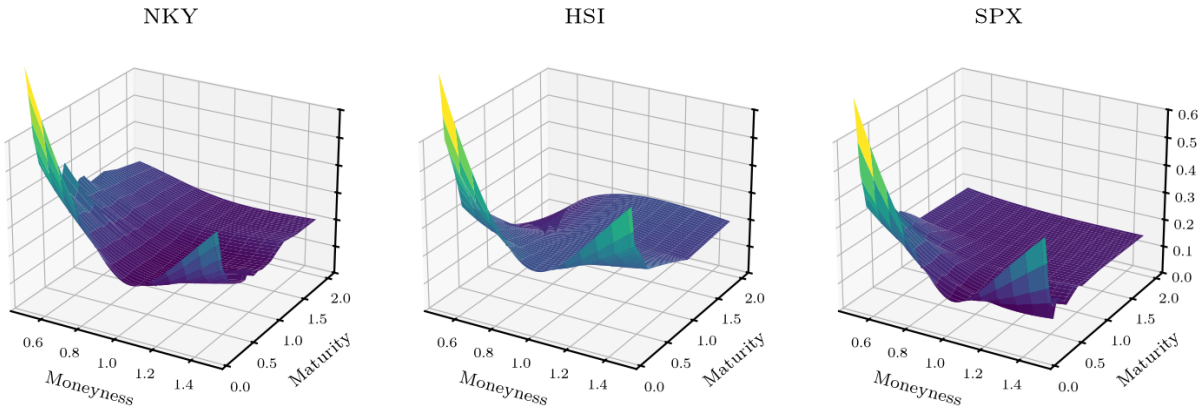


Figure 1: The selected surface, Dupire local volatility interpolated linearly across maturity. The vertical axis is volatility.

5 Monte Carlo Simulation

The correlation matrix of SPX, HSI and NKY is computed from the latest ten years of historical returns. A Cholesky factorization then yields three correlated Gaussian processes $\{Z^k\}$ from three independent Wiener processes $\{W^k\}$. Under the local volatility model, each index follows $dS/S = f\,dt + \sigma_{\text{loc}}\,dW$, where f is the forward rate applying over the step. Applying Ito's lemma to $\ln S$ gives

$$S^k_{t_{j+1}} = S^k_{t_j} \exp\left[\left(\frac{t_{j+1}\,r(t_{j+1}) - t_j\,r(t_j)}{t_{j+1}-t_j} - \tfrac{1}{2}\sigma_j^2\right)(t_{j+1}-t_j) + \sigma_j\sqrt{t_{j+1}-t_j}\,Z^k_j\right]. \tag{5}$$

$$\sigma_j^2 = \sigma^2_{\text{loc}}\left(\ln\frac{S^k_{t_j}}{F^k_{t_j}},\, t_j\right), \qquad F^k_T = S^k_0 \exp\{(r(T)-q(T))T\}.$$

Where the interpolated σ_j^2 is negative, it is replaced by the smallest non-negative value seen so far in the simulation.

5.1 Autocallable pricing

Write $\Pi_m = \prod_{l<m} \mathbf{1}\{R_{t_l} < B_+\}$ for the indicator that no knock-out event has occurred before t_m, and $D_m = e^{-r(t_m)t_m}$ for the discount factor. The value of the note at coupon strike c is the average over the M simulated paths of the sum of three components,

$$V(c) = \frac{1}{M}\sum_{i=1}^{M}(I_i + O_i + G_i). \tag{6}$$

The variable interest accrues at each observation date and stops once the note has knocked out,

$$I = N\sum_{m=1}^{4}\left(\frac{r_+}{2}\,\mathbf{1}\{R_{t_m}\ge c\} + \frac{r_-}{2}\,\mathbf{1}\{R_{t_m}<c\}\right)\Pi_m D_m.$$

The knock-out redemption pays the denomination at the first observation date at which the barrier is breached, at $\tau = \min\{t_m : R_{t_m} \ge B_+\}$,

$$O = N\,\mathbf{1}\{\tau<\infty\}\,e^{-r(\tau)\tau}.$$

The final redemption applies only to paths that never knocked out, and is reduced by the terminal reference price only if the knock-in barrier was breached at some point,

$$G = N\,\mathbf{1}\{\tau=\infty\}\left(\mathbf{1}\{\min_t R_t \le B_-\}\min(1,R_T) + \mathbf{1}\{\min_t R_t > B_-\}\right)D_4.$$

6 Coupon Strike Optimization

We calibrate the coupon strike at which the note prices at 98% of its issue price by employing the bisection algorithm of Appendix A. It terminates when $|V(c) - 0.98N| \le$ USD 0.1, and returns no result if the algorithm exhausts 100 iterations.

At the contractual rates of $r_+ = 2\%$ and $r_- = 0.01\%$ the search returns no result. We attribute the failure to the level of interest rates. The two-year USD rate was 4.83%, so a coupon of 2% falls well below the rate at which the note is discounted. Since the contractual rates admit no solution, we treat the two rates as free and identify the pairs that do. Table 2 reports the coupon strike over a grid of maximum and minimum interest. At a fixed price, the coupon strike increases with both the maximum and minimum interest, as the structure implies.

Maximum	Minimum interest, basis points														
interest, %	1.0	8.1	15.1	22.2	29.3	36.4	43.4	50.5	57.6	64.6	71.7	78.8	85.9	92.9	100.0
2.00	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
3.29	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
4.57	–	–	–	–	–	–	–	–	–	–	–	–	–	–	–
5.86	0.702	0.703	0.704	0.706	0.707	0.708	0.709	0.711	0.712	0.713	0.715	0.716	0.718	0.719	0.720
7.14	0.798	0.799	0.800	0.802	0.803	0.804	0.805	0.807	0.809	0.810	0.812	0.813	0.815	0.816	0.818
8.43	0.846	0.847	0.849	0.850	0.851	0.853	0.854	0.856	0.857	0.859	0.861	0.862	0.864	0.865	0.867
9.71	0.878	0.879	0.880	0.882	0.883	0.884	0.886	0.887	0.889	0.890	0.892	0.893	0.895	0.896	0.898
11.00	0.901	0.902	0.903	0.905	0.906	0.907	0.909	0.910	0.911	0.913	0.914	0.915	0.917	0.918	0.920
12.29	0.918	0.920	0.921	0.922	0.923	0.925	0.926	0.927	0.929	0.930	0.931	0.933	0.934	0.936	0.937
13.57	0.933	0.934	0.935	0.937	0.938	0.939	0.940	0.941	0.943	0.944	0.946	0.947	0.948	0.950	0.952
14.86	0.945	0.946	0.948	0.949	0.950	0.951	0.953	0.954	0.955	0.956	0.958	0.959	0.961	0.962	0.964
16.14	0.956	0.957	0.958	0.959	0.961	0.962	0.963	0.965	0.966	0.967	0.969	0.970	0.971	0.972	0.974
17.43	0.965	0.966	0.968	0.969	0.970	0.971	0.972	0.973	0.975	0.976	0.977	0.979	0.980	0.981	0.983
18.71	0.973	0.974	0.976	0.977	0.978	0.979	0.980	0.981	0.983	0.984	0.985	0.986	0.988	0.989	0.991
20.00	0.981	0.982	0.983	0.984	0.985	0.986	0.987	0.989	0.990	0.991	0.993	0.994	0.995	0.997	0.998

Table 2: Coupon strike that prices the note at 98% of face. A dash marks a pair of coupon rates for which the search returns no result.

7 Conclusion

In this project, we fitted implied and local volatility surfaces on the Nikkei 225, the S&P 500 and the Hang Seng Index. Of the four local variance surfaces, the Dupire local variance interpolated linearly across maturity left the smallest part of the traded moneyness window with negative variance, and we used it for the simulation. We then simulated the three indices jointly and valued a multi-asset autocallable note whose coupon, knock-out and knock-in all reference the laggard index. At the contractual coupon rates of 2% and 0.01%, no coupon strike prices the note at 98% of issue, because a coupon well below the discount rate leaves the note worth less than the target at any strike. Treating the two rates as free, the coupon strike increases with both the maximum and the minimum interest.

It should be noted that the last option price from Bloomberg may not be a reliable input. The last trade may fall well before the snapshot, so the option price, the time to maturity and the spot price are not contemporaneous, which introduces noise into the surface fit.

References

Dupire, B. (1994). Pricing with a smile. *Risk*, 7, 18–20.

Gatheral, J. (2002). Lecture 1: Stochastic volatility and local volatility. Case Studies in Financial Modelling Course Notes, Courant Institute of Mathematical Sciences, Fall Term.

A Bisection

For f continuous on $[a,b]$ with $f(a)f(b)<0$, the intermediate value theorem gives a root $x^*\in(a,b)$. Set $x_l=(a_l+b_l)/2$. If $|f(x_l)|<\varepsilon$ stop, otherwise replace $b_{l+1}=x_l$ when $f(x_l)>0$ and $a_{l+1}=x_l$ when $f(x_l)<0$, and stop when $|b_l-a_l|<\eta$. The bracket halves at each step, so $b_l-a_l=2^{-l}(b-a)$ and $|x_l-x^*|\le 2^{-(l+1)}(b-a)$, giving linear convergence at rate one half.

B Cubic spline

On knots $x_0<\cdots<x_K$ with values f_k, write s_k for the cubic on $[x_k,x_{k+1}]$, $m_k=s''(x_k)$ and $h_k=x_{k+1}-x_k$. Requiring continuity of s, s' and s'' at the interior knots gives

$$h_{k-1}m_{k-1}+2(h_{k-1}+h_k)m_k+h_km_{k+1}=6\left(\frac{f_{k+1}-f_k}{h_k}-\frac{f_k-f_{k-1}}{h_{k-1}}\right),$$

for $k=1,\dots,K-1$. The implementation closes the system by the not-a-knot conditions, which require s''' to be continuous at x_1 and x_{K-1},

$$\frac{m_1-m_0}{h_0}=\frac{m_2-m_1}{h_1},\qquad \frac{m_{K-1}-m_{K-2}}{h_{K-2}}=\frac{m_K-m_{K-1}}{h_{K-1}}.$$

These two conditions determine $\{m_k\}$ uniquely, and the coefficients of s_k follow from m_k and f_k.

C The Dupire equation

In the forward price the diffusion is $dF=\sigma_{\mathrm{loc}}(F,t)F\,dW$. Write $\varphi(F,T)$ for the density of F_T. The undiscounted call is

$$C(K,T)=\int_K^\infty (F-K)\,\varphi(F,T)\,dF,$$

$$\frac{\partial C}{\partial K}=-\int_K^\infty \varphi\,dF,\qquad \frac{\partial^2 C}{\partial K^2}=\varphi(K,T).$$

The density satisfies the Fokker-Planck equation $\partial_T\varphi=\frac12\partial_F^2(\sigma_{\mathrm{loc}}^2F^2\varphi)$. Therefore,

$$\frac{\partial C}{\partial T}=\int_K^\infty (F-K)\,\partial_T\varphi\,dF=\frac12\int_K^\infty (F-K)\,\partial_F^2\big(\sigma_{\mathrm{loc}}^2F^2\varphi\big)\,dF.$$

Integrating by parts twice,

$$\frac{\partial C}{\partial T}=\frac12\,\sigma_{\mathrm{loc}}^2(K,T)\,K^2\,\varphi(K,T)=\frac12\,\sigma_{\mathrm{loc}}^2(K,T)\,K^2\,\frac{\partial^2 C}{\partial K^2},$$

which rearranges to the equation of Section 3.3.

D Gatheral's mapping

For $w=\sigma_{\mathrm{BS}}^2T$, write $C(y,T)=C_{\mathrm{BS}}(y,w(y,T))$ for the market price, which depends on maturity only through w. The Dupire equation of Appendix C gives

$$\frac{\partial C}{\partial T}=\frac{\sigma_{\mathrm{loc}}^2}{2}\left(\frac{\partial^2 C}{\partial y^2}-\frac{\partial C}{\partial y}\right).$$

The Black-Scholes price in w satisfies the identities of Gatheral (2002),

$$\frac{\partial^2 C_{\mathrm{BS}}}{\partial w^2} = \left(-\frac{1}{8} - \frac{1}{2w} + \frac{y^2}{2w^2}\right)\frac{\partial C_{\mathrm{BS}}}{\partial w}, \qquad \frac{\partial^2 C_{\mathrm{BS}}}{\partial y \partial w} = \left(\frac{1}{2} - \frac{y}{w}\right)\frac{\partial C_{\mathrm{BS}}}{\partial w},$$

$$\frac{\partial^2 C_{\mathrm{BS}}}{\partial y^2} - \frac{\partial C_{\mathrm{BS}}}{\partial y} = 2\,\frac{\partial C_{\mathrm{BS}}}{\partial w}.$$

The chain rule through $w(y, T)$ gives

$$\frac{\partial C}{\partial T} = \frac{\partial C_{\mathrm{BS}}}{\partial w}\frac{\partial w}{\partial T}, \qquad \frac{\partial C}{\partial y} = \frac{\partial C_{\mathrm{BS}}}{\partial y} + \frac{\partial C_{\mathrm{BS}}}{\partial w}\frac{\partial w}{\partial y},$$
$$\frac{\partial^2 C}{\partial y^2} = \frac{\partial^2 C_{\mathrm{BS}}}{\partial y^2} + 2\frac{\partial^2 C_{\mathrm{BS}}}{\partial y \partial w}\frac{\partial w}{\partial y} + \frac{\partial^2 C_{\mathrm{BS}}}{\partial w^2}\left(\frac{\partial w}{\partial y}\right)^2 + \frac{\partial C_{\mathrm{BS}}}{\partial w}\frac{\partial^2 w}{\partial y^2}.$$

Therefore,

$$\frac{\partial^2 C}{\partial y^2} - \frac{\partial C}{\partial y} = \frac{\partial C_{\mathrm{BS}}}{\partial w}\left[2 - \frac{2y}{w}\frac{\partial w}{\partial y} + \left(-\frac{1}{8} - \frac{1}{2w} + \frac{y^2}{2w^2}\right)\left(\frac{\partial w}{\partial y}\right)^2 + \frac{\partial^2 w}{\partial y^2}\right].$$

Putting this and $\partial C/\partial T$ into the Dupire equation gives

$$\frac{\partial w}{\partial T} = \sigma_{\mathrm{loc}}^2\left[1 - \frac{y}{w}\frac{\partial w}{\partial y} + \frac{1}{4}\left(-\frac{1}{4} - \frac{1}{w} + \frac{y^2}{w^2}\right)\left(\frac{\partial w}{\partial y}\right)^2 + \frac{1}{2}\frac{\partial^2 w}{\partial y^2}\right].$$

Inverting gives the mapping of Section 3.4.

E Simulation algorithm

Given M paths and J steps, for each path:

1. Evolve the three indices by the recursion of Section 5 and form the reference price R_t at every step.
2. Record the knock-in indicator from the running minimum of R_t over all steps, not only the observation dates.
3. At each observation date t_m, accrue the interest of Section 5.1 if the note is still alive, and record a knock-out at the first date with $R_{t_m} \geq B_+$.
4. Discount each component at its own payment date rather than at maturity, since the knock-out redemption and the coupons are paid when they occur.

Averaging over paths gives $V(c)$, and the coupon strike is the value at which $V(c)$ reaches the target.

F Fitted volatility curves and surfaces

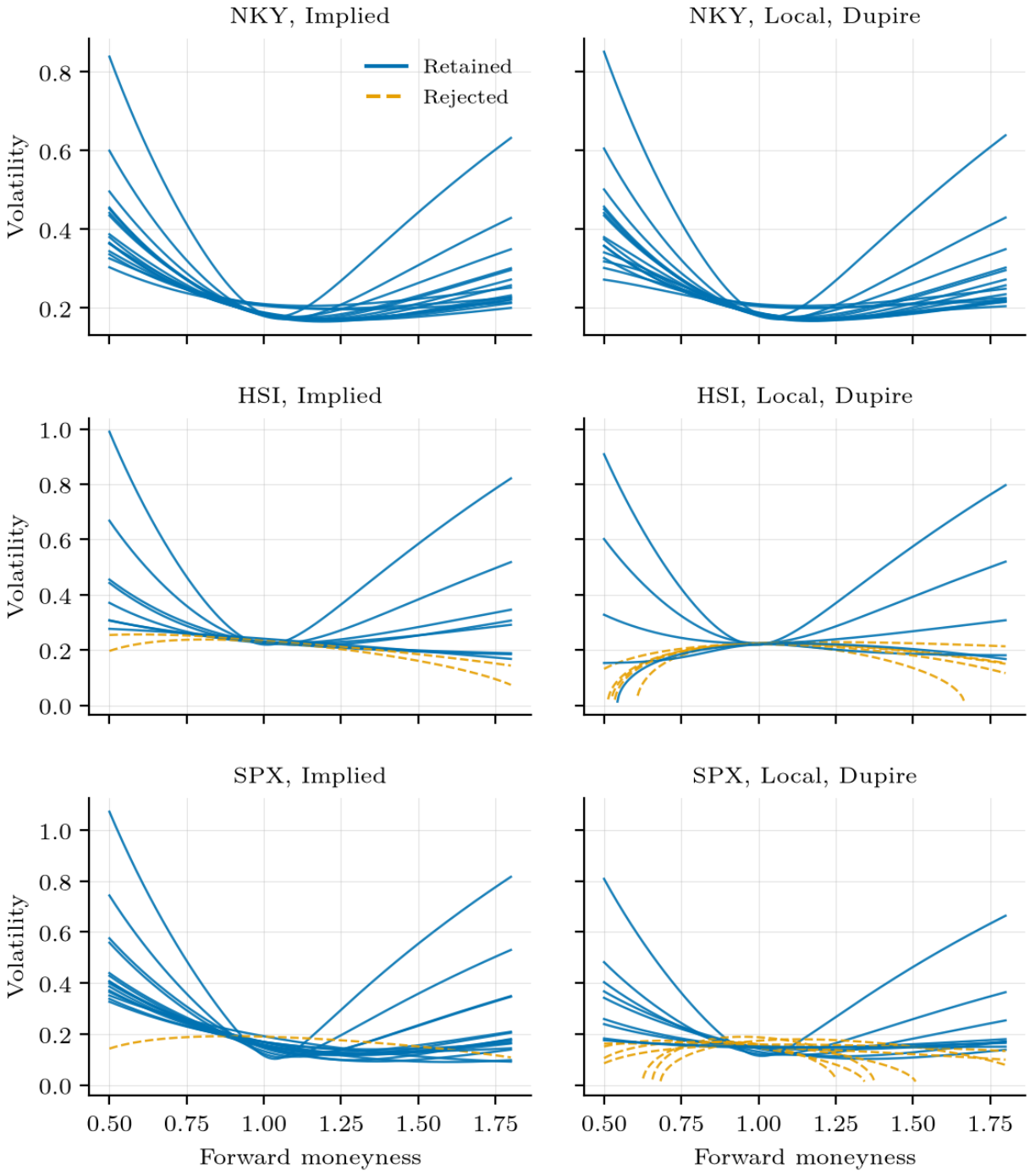


Figure 2: Fitted implied and local volatility curves across forward moneyness.

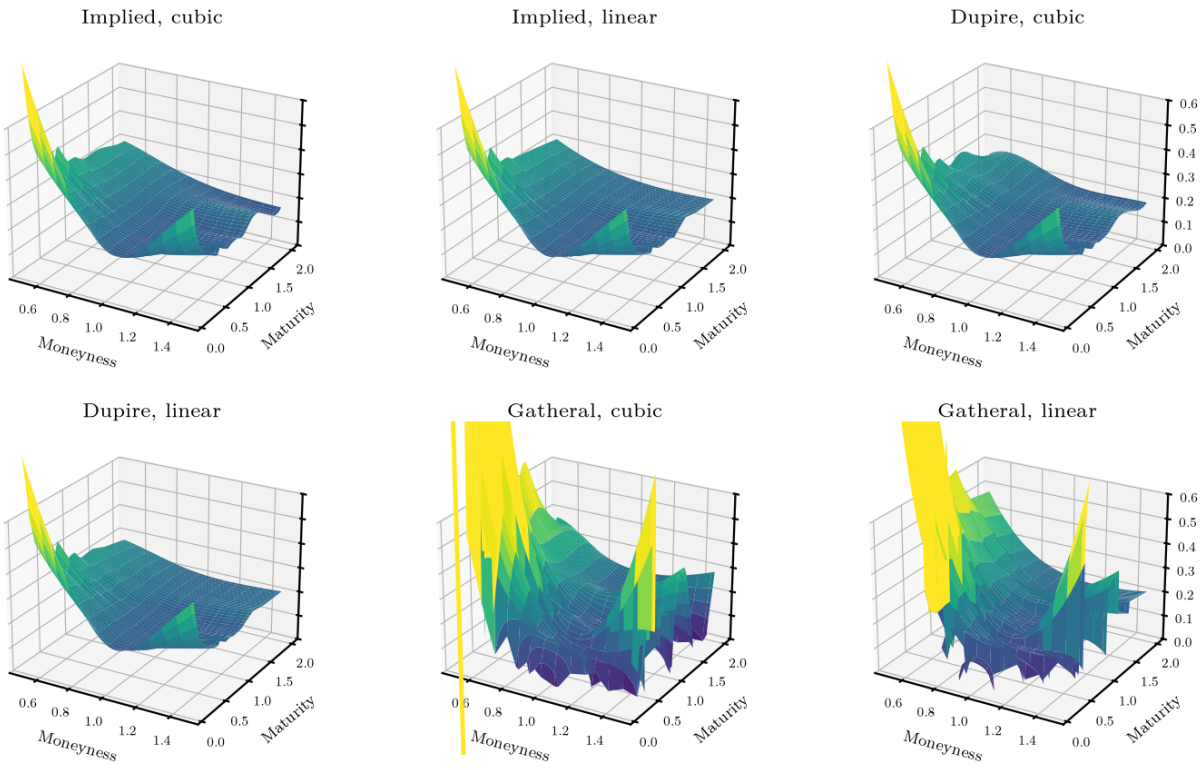


Figure 3: Nikkei 225 fitted surfaces; blank regions carry no real volatility.

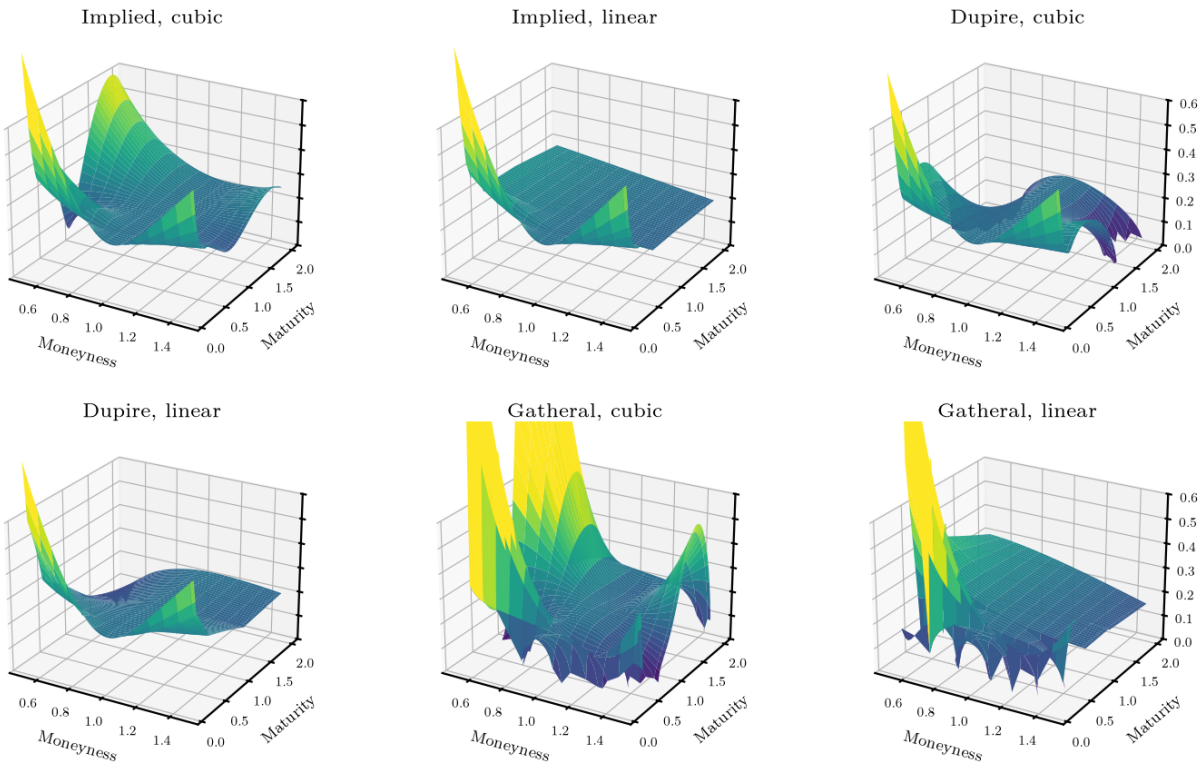


Figure 4: Hang Seng Index fitted surfaces; blank regions carry no real volatility.

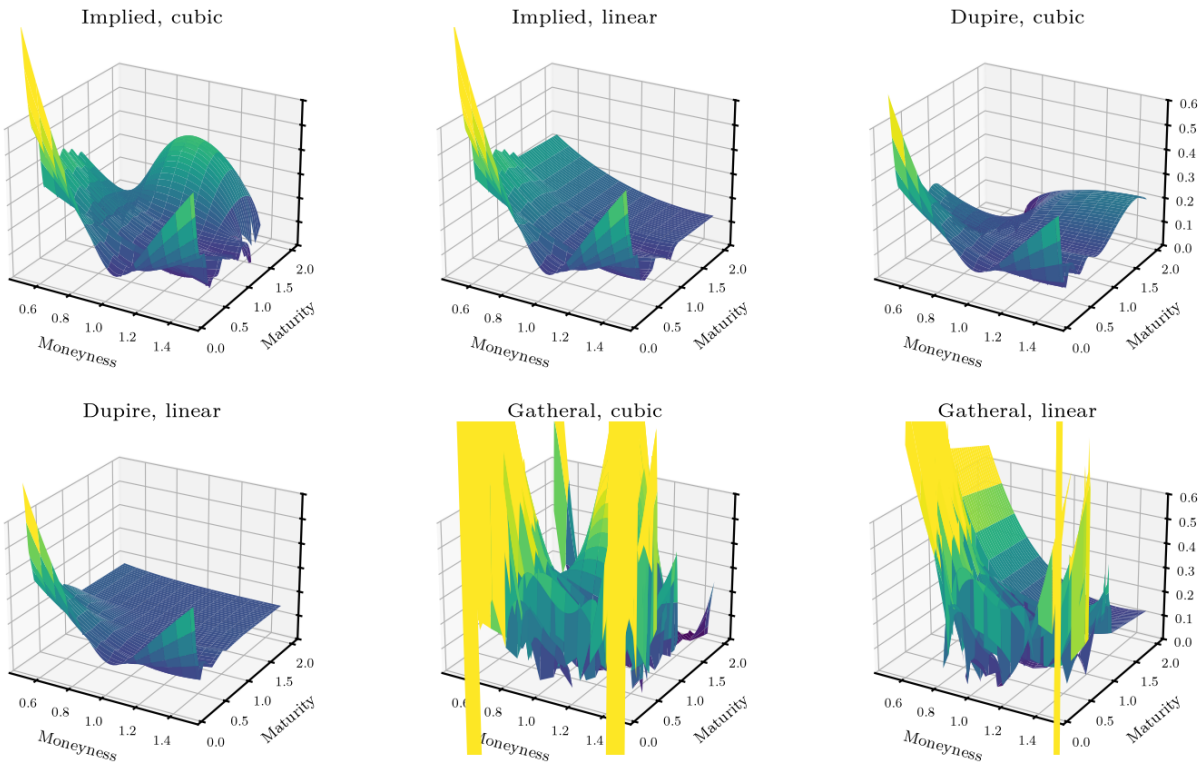


Figure 5: S&P 500 fitted surfaces; blank regions carry no real volatility.